

LAFAYETTE 13 SE, LA, USA

WMO: 747600

Lat: 30.0918N

Lon: 91.8731W

Elev: 35

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	32.8	17.6	13.4	33.7	22.8	17.2	40.2	12.8	67.4	11.8	67.3	1.9	N/A	0.304

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	93.0	77.9	91.4	77.8	90.0	77.5	80.4	88.1	79.9	87.6	79.2	86.8	4.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.9	83.5	77.9	145.0	83.4	77.2	141.8	83.0	44.0	88.1	43.3	87.8	42.7	87.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.4	8.3	DB	24.6	95.9	3.9	2.3	21.8	97.6	19.6	98.9	17.4	100.2	14.6	101.9	(4)
			WB	21.4	82.0	4.3	0.7	18.3	82.5	15.8	82.9	13.3	83.2	10.2	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.2	52.1	55.9	62.2	67.7	74.7	80.1	81.2	81.4	77.9	69.4	60.0	54.5
	DBStd	12.58	10.00	9.48	7.97	6.32	5.05	2.60	2.24	2.66	3.83	7.27	8.87	9.64
	HDD50	232	95	45	10	1	0	0	0	0	0	0	18	63
	HDD65	1471	411	276	147	45	5	0	0	0	0	42	197	348
	CDD50	6862	161	211	388	531	767	903	968	973	838	601	319	202
	CDD65	2629	12	22	61	126	307	453	503	508	388	179	48	22
	CDH74	22405	12	49	222	766	2519	4319	4779	4858	3162	1408	245	66
	CDH80	8164	0	1	6	74	741	1772	1946	2064	1176	366	18	0
Wind	WSAvg	3.3	4.0	4.5	4.1	4.0	3.5	2.8	2.3	2.2	2.4	2.6	3.2	3.7
Precipitation	PrecAvg	61.5	5.6	4.4	3.9	5.0	5.0	6.8	6.0	5.7	5.3	4.7	4.3	4.8
	PrecMax	85.3	13.3	9.1	10.6	11.9	13.5	17.9	9.3	27.9	10.3	13.9	13.8	13.5
	PrecMin	40.4	0.8	0.7	0.3	0.5	0.3	0.9	1.8	1.5	2.0	0.2	1.2	2.3
	PrecStd	11.3	3.6	2.4	2.4	2.9	3.4	3.7	1.9	4.8	2.4	3.9	3.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.7	78.5	80.6	84.4	90.7	95.1	94.3	95.8	92.4	88.9	82.4	78.5
		MCWB	67.6	70.0	69.4	72.3	74.1	76.7	78.0	77.9	77.4	74.6	71.4	71.4
	2%	DB	73.4	75.3	78.5	82.2	87.9	92.0	92.2	93.2	90.2	86.4	79.6	76.0
		MCWB	66.3	68.3	68.7	71.4	73.8	77.2	78.2	78.3	76.8	74.1	70.3	70.2
	5%	DB	70.7	73.0	76.8	80.3	86.1	90.1	90.6	91.4	88.6	84.2	77.0	73.0
		MCWB	65.5	67.8	68.1	70.1	73.7	77.2	78.2	78.6	76.3	72.9	68.6	68.1
	10%	DB	68.0	70.3	74.5	78.5	84.4	88.5	89.0	89.5	86.8	81.8	74.3	70.3
		MCWB	63.5	66.4	67.1	69.2	73.3	77.1	78.0	78.4	75.8	71.3	67.2	66.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	72.0	72.6	75.6	77.4	80.4	81.1	81.5	80.2	79.0	74.9	73.8
		MCDB	73.1	76.3	77.6	81.6	84.6	88.4	88.9	89.3	87.1	84.2	78.3	77.2
	2%	WB	68.9	70.8	71.1	74.2	76.5	79.6	80.2	80.6	79.0	77.6	72.7	71.3
		MCDB	71.7	74.6	75.9	79.5	83.4	87.4	88.0	88.4	86.2	83.4	76.5	74.7
	5%	WB	66.6	69.4	70.0	73.1	75.7	78.9	79.5	80.1	78.2	76.1	70.8	69.7
		MCDB	69.4	73.0	74.5	78.0	82.5	86.6	87.3	88.0	85.4	81.4	74.8	72.6
	10%	WB	63.5	67.9	68.8	72.0	75.0	78.1	78.9	79.4	77.3	74.4	68.7	67.6
		MCDB	66.8	71.0	73.1	76.6	81.7	85.8	86.4	87.2	84.4	79.2	73.1	70.2
Mean Daily Temperature Range	5% DB	MDBR	18.4	17.4	18.4	18.8	17.6	16.6	15.7	16.6	17.3	20.8	20.9	18.6
		MCDBR	17.9	16.9	16.9	18.2	18.3	19.0	17.7	18.7	18.3	19.8	19.9	18.1
		MCWBR	12.6	10.7	9.5	9.1	7.1	6.7	6.5	6.7	7.4	9.9	11.4	12.2
	5% WB	MCDBR	15.0	14.5	15.1	14.7	15.6	16.1	16.1	16.5	16.5	16.7	16.9	16.7
		MCWBR	12.3	11.2	10.0	9.4	6.7	6.8	6.6	6.9	7.4	9.5	11.8	13.5
Clear-Sky Solar Irradiance	taub	0.338	0.355	0.378	0.419	0.447	0.474	0.492	0.486	0.445	0.381	0.358	0.351	
	taud	2.505	2.466	2.389	2.278	2.238	2.207	2.191	2.207	2.289	2.445	2.454	2.474	
	Ebn at Noon	284	289	288	278	270	261	255	256	263	276	274	272	
	Edh at Noon	29	32	37	43	45	46	47	45	41	32	30	28	
All-Sky Solar Radiation	RadAvg	927	1085	1479	1802	1984	1941	1850	1749	1611	1425	1074	829	
	RadStd	90	157	131	130	115	158	134	154	142	164	103	84	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (26 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+144	

Nomenclature: See separate page